

SIMATS ENGINEERING
ASSESSMENT TOOL 1: Data Cleaning Challenge

COURSE NAME: Query Processing for Data Science **COURSE CODE:** DSA0503

COURSE OUTCOME COVERED

CO3: Identify and clean inconsistencies in data by handling missing values, duplicates, outliers, formatting issues, and applying techniques like fuzzy matching and regex.

Assessment Tool Weightage: 35%

Total Marks: 20

Code of Conduct

I _____C. Sai Harshitha (192424346)_____ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: __C. Sai Harshitha_____

Debugging

Criteria	Anomaly Detection (4 Marks)	Cleaning Methods (4 Marks)	Data Transformation (4 Marks)	Analytical Comparison (4 Marks)	Governance Discussion (4 Marks)
Marks					

Total Marks Awarded:

Question 1: E-Commerce Customer Transactions Dataset

An online retail company has collected customer transaction data from multiple sales channels. The management wants to identify the top-performing customer segments and products. However, the dataset contains several quality issues.

CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice	PurchaseDate	City
C101	Raj Kumar	Male	28	Electronics	2	15000 INR	10-01-2025	Chennai
C102	Priya S	Female	35	Fashion	3	1200 INR	11-01-2025	chennai
C103	Arun V	Male	150	Electronics	1	300 USD	12-01-2025	CHENNAI
C104	Kavya R	Female	0	Grocery	-2	800 INR	15-01-2025	Coimbatore
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
C105	Rahul P	Male	42	Fashion	5	2500 INR	18-01-2025	Madurai
NULL	Sneha M	Female	30	Electronics	2	450 USD	05-02-2025	Trichy
C107	Deepak N	Male		Grocery	3	600 INR	07-02-2025	Tiruchirappalli
C108	Meena K	Female	27	Fashion	1	1800 INR	2025-13-01	Salem
C109	John D	Male	31	Electronics	4	250 EUR	15-02-2025	Chennai

Data Issues

- Missing Customer IDs
- Duplicate transactions
- Negative quantities
- Unit prices recorded in different currencies
- Age values such as 0, 150, and blanks
- Inconsistent city names (e.g., Chennai, chennai, CHENNAI)
- Incorrect purchase dates

Tasks

1. Identify and classify all data quality issues. (4 Marks)

Issue	Type
Missing Customer ID	Missing value
Duplicate C105 transaction	Duplicate
Negative quantity (-2)	Invalid value
Different currencies (INR, USD, EUR)	Inconsistent format
Age 0 and 150	Invalid value / outlier
Missing age	Missing value
Chennai, chennai, CHENNAI	Inconsistent category
Trichy and Tiruchirappalli	Inconsistent city names
Invalid date 2025-13-01	Invalid value

2. Design a cleaning strategy for each issue. (4 Marks)

- Fill or remove **missing Customer IDs**.
- Remove **duplicate transactions**.
- Correct or remove **negative quantities**.
- Convert all currencies to **INR**.
- Correct invalid **age values**.
- Handle missing ages.
- Use one standard name for each **city**.
- Convert all dates into the same format.
- Remove or correct invalid dates.

3. Perform data standardization and transformation. (4 Marks)

Create a CSV from the table given in the question and then use **Pandas read_csv()**

Python Code:

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
df = pd.read_csv("ecommerce.csv")
df = df.drop_duplicates()
df["City"] = df["City"].replace({
    "chennai": "Chennai",
    "CHENNAI": "Chennai",
    "Trichy": "Tiruchirappalli"
})
df.loc[(df["Age"] < 1) | (df["Age"] > 120), "Age"] = pd.NA
df.loc[df["Quantity"] < 0, "Quantity"] = pd.NA
df["PurchaseDate"] = pd.to_datetime(
    df["PurchaseDate"], errors="coerce", dayfirst=True
)
print(df)
```

Output:

	CustomerID	CustomerName	Gender	Age	ProductCategory	Quantity	UnitPrice
0	C101	Raj Kumar	Male	28.0	Electronics	2.0	15000
1	C102	Priya S	Female	35.0	Fashion	3.0	1200
2	C103	Arun V	Male	NaN	Electronics	1.0	300
3	C104	Kavya R	Female	NaN	Grocery	NaN	800
4	C105	Rahul P	Male	42.0	Fashion	5.0	2500
6	NaN	Sneha M	Female	30.0	Electronics	2.0	450
7	C107	Deepak N	Male	NaN	Grocery	3.0	600
8	C108	Meena K	Female	27.0	Fashion	1.0	1800
9	C109	John D	Male	31.0	Electronics	4.0	250

	Currency	PurchaseDate	City
0	INR	2025-01-10	Chennai
1	INR	2025-01-11	Chennai
2	USD	2025-01-12	Chennai
3	INR	2025-01-15	Coimbatore
4	INR	2025-01-18	Madurai
6	USD	2025-02-05	Tiruchirappalli
7	INR	2025-02-07	Tiruchirappalli
8	INR	NaT	Salem
9	EUR	2025-02-15	Chennai

4. Analyze how cleaning affects total revenue calculations. (4 Marks)

Python Code:

```
▶ rates = {  
    "INR": 1,  
    "USD": 85,  
    "EUR": 90  
}  
df["Price_INR"] = df["UnitPrice"] * df["Currency"].map(rates)  
df["Revenue"] = df["Quantity"] * df["Price_INR"]  
print("Revenue Table:")  
display(df[["CustomerID", "ProductCategory", "Quantity",  
            "UnitPrice", "Currency", "Price_INR", "Revenue"]])  
print("Total Revenue:", df["Revenue"].sum())
```

Output:

	CustomerID	ProductCategory	Quantity	UnitPrice	Currency	Price_INR	Revenue
0	C101	Electronics	2.0	15000	INR	15000	30000.0
1	C102	Fashion	3.0	1200	INR	1200	3600.0
2	C103	Electronics	1.0	300	USD	25500	25500.0
3	C104	Grocery	NaN	800	INR	800	NaN
4	C105	Fashion	5.0	2500	INR	2500	12500.0
6	NaN	Electronics	2.0	450	USD	38250	76500.0
7	C107	Grocery	3.0	600	INR	600	1800.0
8	C108	Fashion	1.0	1800	INR	1800	1800.0
9	C109	Electronics	4.0	250	EUR	22500	90000.0

Total Revenue: 241700.0

5. Generate a report discussing business risks if the data were not cleaned (4 Marks)

If the data is not cleaned:

- Revenue may be wrong.
- Sales may be counted twice.
- Customer information may be incorrect.
- Product performance may be wrong.
- City-wise sales analysis may be incorrect.
- Management may make wrong business decisions.

Question 2: Hospital Patient Records Dataset

A hospital wants to analyze patient admissions and treatment effectiveness. Data were entered manually by different departments, leading to inconsistencies.

PatientID	Name	Age	Gender	AdmissionDate	DischargeDate	Diagnosis	TreatmentCost	DoctorID
P1001	Ravi Kumar	45	M	10-01-2025	15-01-2025	Diabetes	25000	D101
P1002	Priya S	38	F	12-01-2025	11-01-2025	Hypertension	18000	D102
P1003	Arun V	125	M	13-01-2025	20-01-2025	diabetes	22000	D101
NULL	Kavya R	29	F	15-01-2025	17-01-2025	Diabtes	15000	D103
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1005	Rahul P	51	M	18-01-2025	25-01-2025	Asthma	NULL	D104
P1006	Sneha M	34	F	15-01-2025	18-01-2025	Hypertension	12000	D102
P1007	Deepak N		M	01-02-2025	05-02-2025	Asthma	10000	D104
P1008	Meena K	26	F	10-02-2025	15-02-2025	HYPERTENSION	13500	D102
P1009	John D	40	M	12-02-2025	18-02-2025	Diabetes	20000	D101

Data Issues

- Missing patient IDs
- Duplicate patient records
- Discharge dates before admission dates
- Null treatment costs
- Inconsistent diagnosis names
- Multiple formats of dates
- Unrealistic patient ages (>120)

Tasks

1. Detect all inconsistencies and anomalies. (4 Marks)

Issue	Type
Missing Patient ID	Missing value
Duplicate P1005 record	Duplicate
Discharge date before admission date	Invalid value
Missing treatment cost	Missing value
Diabetes, diabetes, Diabtes	Inconsistent diagnosis
Different date formats	Inconsistent format
Age 125	Invalid value

2. Develop rules for handling missing and invalid values. (4 Marks)

Missing Patient ID: Check the original hospital record and fill it if possible.

Duplicate records: Remove the duplicate record.

Wrong discharge date: Check the original record and correct it.

Missing treatment cost: Fill it from the hospital billing record. If unavailable, keep it as missing.

Invalid age: Check the original record and correct it. Otherwise, mark it as missing.

Invalid dates: Correct them and use one standard date format.

3. Create a process to standardize diagnosis names. (4 Marks)

Python Code:

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
df = pd.read_csv("hospital.csv")
df = df.drop_duplicates()
df["Diagnosis"] = df["Diagnosis"].replace({
    "diabetes": "Diabetes",
    "Diabtes": "Diabetes",
    "HYPERTENSION": "Hypertension"
})
df.loc[(df["Age"] < 1) | (df["Age"] > 120), "Age"] = pd.NA
df["AdmissionDate"] = pd.to_datetime(
    df["AdmissionDate"], errors="coerce", dayfirst=True
)
df["DischargeDate"] = pd.to_datetime(
    df["DischargeDate"], errors="coerce", dayfirst=True
)
print(df)
```

Output:

	PatientID	Name	Age	Gender	AdmissionDate	DischargeDate
0	P1001	Ravi Kumar	45.0	M	2025-01-10	2025-01-15
1	P1002	Priya S	38.0	F	2025-01-12	2025-01-11
2	P1003	Arun V	NaN	M	2025-01-13	2025-01-20
3	NaN	Kavya R	29.0	F	2025-01-15	2025-01-17
4	P1005	Rahul P	51.0	M	2025-01-18	2025-01-25
6	P1006	Sneha M	34.0	F	2025-01-15	2025-01-18
7	P1007	Deepak N	35.0	M	2025-02-01	2025-02-05
8	P1008	Meena K	26.0	F	2025-02-10	2025-02-15
9	P1009	John D	40.0	M	2025-02-12	2025-02-18

	Diagnosis	TreatmentCost	DoctorID
0	Diabetes	25000.0	D101
1	Hypertension	18000.0	D102
2	Diabetes	22000.0	D101
3	Diabetes	15000.0	D103
4	Asthma	NaN	D104
6	Hypertension	12000.0	D102
7	Asthma	10000.0	D104
8	Hypertension	13500.0	D102
9	Diabetes	20000.0	D101

4. Evaluate how cleaning changes average treatment cost and hospital stay duration. (4 Marks)

Python Code:

```
df["StayDays"] = (
    df["DischargeDate"] - df["AdmissionDate"]
).dt.days
avg_cost = df["TreatmentCost"].mean()
avg_stay = df["StayDays"].mean()
print("Average Treatment Cost:", avg_cost)
print("Average Hospital Stay:", avg_stay, "days")
display(df[
    "PatientID", "Diagnosis", "TreatmentCost",
    "AdmissionDate", "DischargeDate", "StayDays"
])
```

Output:

Average Treatment Cost: 16937.5
Average Hospital Stay: 4.222222222222222 days

	PatientID	Diagnosis	TreatmentCost	AdmissionDate	DischargeDate	StayDays
0	P1001	Diabetes	25000.0	2025-01-10	2025-01-15	5
1	P1002	Hypertension	18000.0	2025-01-12	2025-01-11	-1
2	P1003	Diabetes	22000.0	2025-01-13	2025-01-20	7
3	NaN	Diabetes	15000.0	2025-01-15	2025-01-17	2
4	P1005	Asthma	NaN	2025-01-18	2025-01-25	7
6	P1006	Hypertension	12000.0	2025-01-15	2025-01-18	3
7	P1007	Asthma	10000.0	2025-02-01	2025-02-05	4
8	P1008	Hypertension	13500.0	2025-02-10	2025-02-15	5
9	P1009	Diabetes	20000.0	2025-02-12	2025-02-18	6

5. Recommend data governance measures to prevent future errors. (4 Marks)

The hospital can prevent future errors by:

- Using standard data-entry formats.
- Making Patient ID a required field.
- Preventing duplicate records.
- Using valid age ranges.
- Using one standard date format.
- Using a fixed list of diagnosis names.
- Checking missing values before saving records.
- Regularly checking and cleaning the database.

Question 3: Banking Loan Approval Dataset

A bank uses customer data to determine loan eligibility. The dataset contains errors that may affect risk assessment and approval decisions.

CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
L001	750000	720	300000	Full Time	35	50000	Approved
L002	900K	680	400000	FT	42	100000	Approved
L003	1200000	NULL	600000	fulltime	28	200000	Approved
L004	150000	450	-200000	Part Time	30	50000	Rejected
L005	5000000	790	1000000	Self Employed	16	10000	Approved
L006	450000	350	250000	Unemployed	40	300000	Rejected
L007	75000000	800	500000	Full Time	45	100000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L008	650000	690	350000	FT	33	70000	Approved
L010	NULL	710	300000	Full Time	37	80000	Approved

Data Issues

- Missing credit scores
- Income values recorded in different units
- Negative loan amounts
- Duplicate customer IDs
- Age values below 18
- Employment status entered differently (FT, Full Time, fulltime)
- Extreme outliers in income

Tasks

1. Identify critical issues that could impact loan decisions. (4 Marks)

Issue	Type
Missing credit score	Missing value
Income in different formats	Inconsistent data
Negative loan amount	Invalid value
Duplicate L008 record	Duplicate
Age 16	Invalid value
FT, Full Time, fulltime	Inconsistent data
Very high income values	Outlier

2. Propose cleaning techniques for missing and inconsistent data. (4 Marks)

- Fill missing credit scores using reliable customer records.
- Convert all income values into the same unit and format.
- Remove or correct negative loan amounts.
- Remove duplicate customer records.
- Correct ages below 18.
- Convert FT, Full Time, and fulltime into **Full Time**.
- Check extreme income values before using them.

3. Detect and justify treatment of outliers. (4 Marks)

Python Code:

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
df = pd.read_csv("loan (1).csv")
df["AnnualIncome"] = df["AnnualIncome"].replace("900K", "900000")
df["AnnualIncome"] = pd.to_numeric(df["AnnualIncome"], errors="coerce")
Q1 = df["AnnualIncome"].quantile(0.25)
Q3 = df["AnnualIncome"].quantile(0.75)
IQR = Q3 - Q1
outliers = df[
    (df["AnnualIncome"] < Q1 - 1.5 * IQR) |
    (df["AnnualIncome"] > Q3 + 1.5 * IQR)
]
print("Outliers:")
display(outliers)
```

Output:

Outliers:

	CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
4	L005	5000000.0	790.0	1000000	Self Employed	16	10000	Approved
6	L007	75000000.0	800.0	500000	Full Time	45	100000	Approved

4. Analyze the effect of cleaning on loan approval rates. (4 Marks)

Python Code:

```
df = df.drop_duplicates()
df.loc[df["Age"] < 18, "Age"] = pd.NA
print("Cleaned Dataset:")
display(df)
print("Loan Approval Comparison:")
before = 10
after = len(df)
approved_before = 8
approved_after = (df["LoanStatus"] == "Approved").sum()
print("Approved Before Cleaning:", approved_before)
print("Approved After Cleaning:", approved_after)
print("Approval Rate Before Cleaning:", approved_before / before * 100, "%")
print("Approval Rate After Cleaning:", approved_after / after * 100, "%")
```

Output:

	CustomerID	AnnualIncome	CreditScore	LoanAmount	EmploymentStatus	Age	ExistingDebt	LoanStatus
0	L001	750000.0	720.0	300000	Full Time	35.0	50000	Approved
1	L002	900000.0	680.0	400000	FT	42.0	100000	Approved
2	L003	1200000.0	NaN	600000	fulltime	28.0	200000	Approved
3	L004	150000.0	450.0	-200000	Part Time	30.0	50000	Rejected
4	L005	5000000.0	790.0	1000000	Self Employed	NaN	10000	Approved
5	L006	450000.0	350.0	250000	Unemployed	40.0	300000	Rejected
6	L007	75000000.0	800.0	500000	Full Time	45.0	100000	Approved
7	L008	650000.0	690.0	350000	FT	33.0	70000	Approved
9	L010	NaN	710.0	300000	Full Time	37.0	80000	Approved

Loan Approval Comparison:
Approved Before Cleaning: 8
Approved After Cleaning: 7
Approval Rate Before Cleaning: 80.0 %
Approval Rate After Cleaning: 77.77777777777779 %

5. Discuss ethical implications of incorrect data cleaning decisions. (4 Marks)

Incorrect data cleaning can:

- Reject eligible customers.
- Approve risky customers.
- Treat customers unfairly.
- Create biased loan decisions.
- Cause financial loss to customers and the bank.

Question 4: Smart City Traffic Monitoring Dataset

A smart city project collects traffic sensor data to optimize traffic signals. Sensor malfunctions and transmission errors have affected data quality.

SensorID	Timestamp	VehicleCount	Average Speed	TrafficSignalStatus	RoadName	Weather Condition
S101	01-01-2025 08:00	120	45	Green	Anna Salai	Sunny
S102	01-01-2025 08:05	135	52	Red	anna salai	Sunny
S103	01-01-2025 08:10	-50	48	Green	ANNA SALAI	Cloudy
S104	01-01-2025 08:15	140	350	Yellow	GST Road	Rainy
S105	NULL	110	42	Green	GST ROAD	Sunny
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S106	01-01-2025 08:25	130	46	Red	OMR	NULL
S107	01-01-2025 08:30	150	55	Green	Old Mahabalipuram Road	Sunny
S10A	01-01-2025 08:35	145	53	Green	OMR	Cloudy
S109	01-01-2025 08:40	138	50	Green	GST Road	Rainy

Data Issues

- Missing timestamps
- Duplicate sensor readings
- Vehicle counts recorded as negative numbers
- Average speed exceeding 300 km/h
- Missing weather information
- Inconsistent road naming conventions
- Sensor IDs with formatting errors

Tasks

1. Assess the quality of the traffic dataset. (4 Marks)

Issue	Type
Missing timestamp	Missing value
Duplicate S106 reading	Duplicate
Vehicle count -50	Invalid value
Speed 350 km/h	Outlier
Missing weather	Missing value
Anna Salai, anna salai, ANNA SALAI	Inconsistent name
Old Mahabalipuram Road	Inconsistent name
Sensor ID S10A	Formatting error

2. Recommend cleaning techniques for time-series data. (4 Marks)

- Fill or remove missing timestamps.
- Remove duplicate sensor readings.
- Replace negative vehicle counts with missing values.
- Check and correct unrealistic speeds.
- Fill missing weather information if possible.
- Standardize road names.
- Correct sensor ID formats.
- Sort the data according to timestamp.

3. Design rules to identify faulty sensor readings. (4 Marks)

Python Code:

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
df = pd.read_csv("traffic.csv")
df = df.drop_duplicates()
df.loc[df["VehicleCount"] < 0, "VehicleCount"] = pd.NA
df.loc[df["AverageSpeed"] > 300, "AverageSpeed"] = pd.NA
df["Timestamp"] = pd.to_datetime(
    df["Timestamp"], errors="coerce"
)
print(df)
```

Output:

	SensorID	Timestamp	VehicleCount	AverageSpeed	\
0	S101	2025-01-01 08:00:00	120.0	45.0	
1	S102	2025-01-01 08:05:00	135.0	52.0	
2	S103	2025-01-01 08:10:00	NaN	48.0	
3	S104	2025-01-01 08:15:00	140.0	NaN	
4	S105	NaT	110.0	42.0	
5	S106	2025-01-01 08:25:00	130.0	46.0	
7	S107	2025-01-01 08:30:00	150.0	55.0	
8	S10A	2025-01-01 08:35:00	145.0	53.0	
9	S109	2025-01-01 08:40:00	138.0	50.0	

	TrafficSignalStatus	RoadName	WeatherCondition
0	Green	Anna Salai	Sunny
1	Red	anna salai	Sunny
2	Green	ANNA SALAI	Cloudy
3	Yellow	GST Road	Rainy
4	Green	GST ROAD	Sunny
5	Red	OMR	NaN
7	Green	Old Mahabalipuram Road	Sunny
8	Green	OMR	Cloudy
9	Green	GST Road	Rainy

4. Evaluate how cleaning affects congestion analysis. (4 Marks)

Python Code:

```
df["Congestion"] = (  
    (df["VehicleCount"] > 130) &  
    (df["AverageSpeed"] < 40)  
)  
print("Congested Readings:", df["Congestion"].sum())
```

Output:

Congested Readings: 0

5. Suggest automated monitoring techniques for future data collection. (4 Marks)

- Use automatic range checks for speed and vehicle count.
- Detect duplicate sensor readings automatically.
- Generate alerts for missing data.
- Validate sensor IDs automatically.
- Monitor unusual changes in sensor values.
- Use real-time data-quality dashboards.

Question 5: Global Sales Performance Dataset

A multinational company has merged sales data from five countries. Management wants to evaluate regional performance and forecast future sales. The merged dataset contains severe data quality problems.

Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
Asia	India	Raj Kumar	Laptop	50000	INR	10-01-2025	4
Asia	India	Rajkumar	Laptop	55000	INR	10-01-2025	5
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
Europe	Germany	Anna M	Laptop	1200	EUR	12-01-2025	4
North America	USA	John D	Laptop	1500	USD	01-15-2025	6
Asia	India	Priya S	Mobile	NULL	INR	18-01-2025	3
Europe	France	Pierre L	Ordinateur Portable	1300	EUR	20-01-2025	4
Asia	India	Rahul P	Mobile	9999999	INR	25-01-2025	5
UK	UK	David T	Laptop	1100	GBP	28-01-2025	2
Asia	India	Priya S	Smartphone	35000	INR	30-01-2025	0

Data Issues

- Multiple currencies (USD, EUR, INR, GBP)
- Missing sales amounts
- Duplicate regional entries
- Customer ratings outside the scale (1–5)
- Different date formats across countries
- Product names in multiple languages
- Salespersons recorded with spelling variations
- Extreme sales outliers due to data-entry errors

Tasks

1. Develop a comprehensive data-cleaning framework for multinational datasets. (4 Marks)

Step	Action
Identify errors	Find missing, duplicate and invalid data
Remove duplicates	Remove repeated sales records
Handle missing values	Fill or remove missing sales amounts
Standardize currencies	Convert all currencies to one currency
Standardize dates	Use one date format
Standardize products	Use one common product name
Standardize salesperson names	Correct spelling variations
Check ratings	Keep ratings between 1 and 5
Handle outliers	Check unusually high sales values
Validate data	Check the final dataset for errors

2. Design methods to standardize currencies, products, and dates. (4 Marks)
3. Handle missing values and outliers using appropriate analytical techniques. (4 Marks)
4. Compare regional sales rankings before and after cleaning. (4 Marks)

Python Code:

```
from google.colab import files
uploaded = files.upload()
import pandas as pd
df = pd.read_csv("sales.csv")
before = df.groupby("Region")["SalesAmount"].sum().sort_values(ascending=False)
df = df.drop_duplicates()
df["Product"] = df["Product"].replace({
    "Ordinateur Portable": "Laptop"
})
df["SalesPerson"] = df["SalesPerson"].replace({
    "Rajkumar": "Raj Kumar"
})
df["SalesDate"] = pd.to_datetime(
    df["SalesDate"], errors="coerce", dayfirst=True
)
df.loc[(df["CustomerRating"] < 1) |
       (df["CustomerRating"] > 5), "CustomerRating"] = 5

df["SalesAmount"] = pd.to_numeric(
    df["SalesAmount"], errors="coerce"
)
df["SalesAmount"] = df["SalesAmount"].fillna(
    df["SalesAmount"].median()
)
Q1 = df["SalesAmount"].quantile(0.25)
Q3 = df["SalesAmount"].quantile(0.75)
IQR = Q3 - Q1
df.loc[
    (df["SalesAmount"] < Q1 - 1.5 * IQR) |
    (df["SalesAmount"] > Q3 + 1.5 * IQR),
    "SalesAmount"
] = df["SalesAmount"].median()
after = df.groupby("Region")["SalesAmount"].sum().sort_values(
    ascending=False
)
print("Cleaned Dataset:")
display(df)
print("Regional Sales Before Cleaning:")
display(before)
print("Regional Sales After Cleaning:")
display(after)
```

Output:

Cleaned Dataset:

	Region	Country	SalesPerson	Product	SalesAmount	Currency	SalesDate	CustomerRating
0	Asia	India	Raj Kumar	Laptop	50000.0	INR	2025-01-10	4
1	Asia	India	Raj Kumar	Laptop	55000.0	INR	2025-01-10	5
2	Europe	Germany	Anna M	Laptop	1200.0	EUR	2025-01-12	4
4	North America	USA	John D	Laptop	1500.0	USD	NaT	5
5	Asia	India	Priya S	Mobile	18250.0	INR	2025-01-18	3
6	Europe	France	Pierre L	Laptop	1300.0	EUR	2025-01-20	4
7	Asia	India	Rahul P	Mobile	18250.0	INR	2025-01-25	5
8	UK	UK	David T	Laptop	1100.0	GBP	2025-01-28	2
9	Asia	India	Priya S	Smartphone	35000.0	INR	2025-01-30	5

Regional Sales Before Cleaning:		Regional Sales After Cleaning:	
SalesAmount		SalesAmount	
Region		Region	
Asia	10139999.0	Asia	176500.0
Europe	3700.0	Europe	2500.0
North America	1500.0	North America	1500.0
UK	1100.0	UK	1100.0
dtype: float64		dtype: float64	

5. Prepare an executive report explaining how data quality influences strategic business decisions. (4 Marks)
- Poor data quality can lead to incorrect business decisions. Duplicate records can increase sales figures, missing values can reduce the reported sales, and different currencies can make regional comparisons incorrect. Invalid ratings and product names can also affect customer and product analysis.
- Therefore, the company should clean and standardize its sales data before making decisions about regional performance, products, customers and future sales.